

Theoretical Underpinnings of Computing Education Research – What is the Evidence?

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ABSTRACT

We analyze the Computing Education Research (CER) literature to discover what theories, conceptual models and frameworks recent CER builds on. This gives rise to a broad understanding of the theoretical basis of CER that is useful for researchers working in that area, and has the potential to help CER develop its own identity as an independent field of study.

Our analysis takes in seven years of publications (2005-2011, 308 papers) in three venues that publish long research papers in computing education: the journals ACM Transactions of Computing Education (TOCE) and Computer Science Education (CSEd), and the conference International Computing Education Research Workshop (ICER). We looked at the theoretical background works that are *used or extended* in the papers, not just referred to when describing related work. These background works include theories, conceptual models and frameworks. For each background work we tried to identify the discipline from which it originates, to gain an understanding of how CER relates to its neighboring fields. We also identified theoretical works originating within CER itself, showing that the field is building on its own theoretical works.

Our main findings are that there is a great richness of work on which recent CER papers build; there are no prevailing theoretical

or technical works that are broadly applied across CER; about half the analyzed papers build on no previous theoretical work, but a considerable share of these are building their own theoretical constructions. We discuss the significance of these findings for the whole field and conclude with some recommendations.

Categories and Subject Descriptors

K.3.2 [Computers and education]: Computer and Information Science Education – *computer science education*; A.0 [General]: *conference proceedings*.

Keywords

Classifying publications, computing education, research methods.

1. INTRODUCTION

Computing Education Research (CER) is a relatively new field of investigation, which has emerged from the longer-standing traditions of practice reports and scholarship of learning and teaching in computing education. CER seeks to gain deep understanding of multiple aspects of the teaching and learning processes of various topics in the computing curriculum, and to build generalizable evidence about problems in students' learning and the efficacy of new teaching approaches to solve these problems.

The computing education research tradition is still young, with no firmly established ways of carrying out research. Instead it is often characterized as drawing on the approaches and methods of disciplines with more established traditions of research, such as cognitive psychology, education, and computer science. However, over the past 10 years CER has started forming its own identity as an independent field of study, as can be shown using the six structural criteria proposed by Fensham [7] for the recognition of a new research field. These criteria include, for example, research

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journals, research conferences, and research training, all of which can be found in computing education.

In examining the theories that underlie CER we are applying two of Fensham's intra-research criteria. The first is *conceptual and theoretical development*, which discusses important concepts that can be used to reduce the multiplicity of observations on a more abstract level and theoretical models that present relations between such concepts, and which have some predictive and explanatory power concerning the phenomena of interest. The second is *progression*, which discusses whether the field is using and refining previously developed concepts and theoretical models "...to expand and deepen our appreciation and understanding of ... education and its situation of occurrence" [7]. We have chosen these criteria because they are the ones that will be most readily in evidence when examining the literature.

Extensive use of existing theory is a normal part of research in the natural and human sciences, but not in computer science, where theories are few and the constructive, design and formal approaches dominate. We therefore consider it likely that for CER researchers, most of whom have computer science as their background, it is not straightforward to apply theories in the same way as in the disciplines from which CER draws.

Beyond looking for evidence that theories are being used, we are interested in whether we can find theoretical development work that originates from CER itself. CER investigates the challenges and processes in teaching and learning computing. It seems likely that this requires a theoretical understanding that is inherently related to the concepts and processes of computing itself. Neither general educational theories nor theories from neighboring fields such as Engineering Education Research can be used to explain, for example, how students understand programming concepts. Our research community needs to define its own discipline-based understanding of such issues – although of course we would expect this understanding to build where appropriate on more general theories.

The research reported here emerges from our belief that a more holistic understanding of our own field can support the whole CER community in building its identity as an independent area of research. Thus we hope to address questions such as: On what kind of work does the CER community build its current work? How can the community improve its theoretical understanding of teaching and learning processes in the computing domain? What actions could be carried out to build stronger theoretical foundations in future CER work? Addressing these questions entails an analysis of the current state of art in the field, and this research takes a step in that direction.

An overview of the theoretical foundations of CER is useful for many actors in the field, including researchers, teachers, managers, and reviewers. Revealing the richness of perspectives supports and helps the creation of new research directions and points of view, and helps to grasp the big picture of what research is being conducted in computing education. It is highly useful in research training for the field. It also helps to strengthen the identity of CER as an independent field of research by providing a picture of where we are now as a research community and where we might proceed in the future, exploiting and enhancing theoretical models that can be used for more powerful explanatory and predictive purposes. A similar argument has been presented in mathematics education research: *"One question that repeatedly*

confronts the field is whether or not mathematics education research is a scientific discipline akin to the hard sciences. If so, we need to consider the important role of theory building and theory usage in mathematics education research." [29].

In this paper we present the results of a survey of 308 research papers published in the journals *Computer Science Education*, and *Transactions on Computing Education*, and the *International Computing Education Research Workshop* in the seven years from 2005 to 2011. Our research questions are:

- What theoretical constructs are used in CER?
- From what disciplines do these constructs come?
- Are there indications that CER is building on its own theoretical constructs?

2. RELATED WORK

In recent years there have been a number of efforts to analyze the CER literature to form an overview of the research that is being conducted. This work has focused mainly on two aspects of research: the content of the research and the research process.

With regard to research content, various subareas of CER have been identified by Fincher and Petre [8] and by Pears et al. [21], using different criteria for partitioning the field. Simon et al. [25, 27] presented a more elaborate classification scheme that categorizes research by identifying the curricular context in which the research is conducted, the theme of the research within that context, the scope of the work, and the nature of the research setting. The system has been applied to nearly 600 papers from four computing education conferences and one journal over the years 2000-2007; see Simon [26] and the references in that paper. Other surveys include that of Joy et al. [13], who categorized content, considering particularly where certain kinds of work were published, and that of Kinnunen et al. [14], who looked at the pedagogical focuses of the papers published in ICER.

The second branch in the literature analysis has looked at the research process. Randolph and several other researchers critically reviewed papers in eight CS education publication forums during the years 2000-2005, focusing on the research process: whether human participants were involved, what kind of research setting was used, what data was collected, how it was analyzed, and how the research was reported. They identified many weaknesses in reported research settings [22, 23]. In our own prior work [16] we have sought to characterize research in computing education using a system that includes the theoretical basis of the research.

Sheard et al. [24] touched on both of these branches when reviewing the programming education research papers from six CER conferences from the years 2005-2008. They categorized the papers using Simon's system, but also looked at the research methods used in those papers.

However, through all of these surveys we found very little attention paid to the theoretical basis of the research being reported. We therefore broadened our scope to look at research in other areas of STEM education, to find out what it could tell us about the use of theory in those areas. Again we found many literature surveys examining various aspects of research publications, but few that analyzed the use and application of theories.

In Engineering Education Research (EER) we found only one survey addressing theories. Wankat [31] analyzed 597 papers published in the Journal of Engineering Education in the years 1993-2002. The analysis included the theoretical backgrounds to the work, noting that “a judgment was made as to whether a theory was merely being cited or it was actually being used for design or analysis”. The results showed that an overwhelming majority (84%) of EER papers published during this period did not use any educational theory. The most common theories found were Kolb’s experiential learning, learning styles, and the MBTI indicator, which were used in 1-3% of the papers. This survey is 10 years old, and the situation might well have changed, but we have found no more recent work on the topic.

In Physics Education Research (PER) a recent synthesis report [6] discusses the use of theoretical frameworks in different subareas of PER, and points out the need to build theoretical frameworks for PER. More than a decade earlier, a resource letter [17] presented a large list of papers in various areas of physics education research, and identified a set of example papers with theoretical development in the areas of concept development and problem solving performance.

In Mathematics Education Research (MER), theoretical foundations and the role of theory development have been discussed extensively. Sriraman and English [30] conducted a survey of the use of theories in mathematics education papers, and a Theory of Mathematics Education study group organized many international conferences on the topic in the 1980s and 1990s [29]. Work from MER is discussed in more detail in the next section.

3. WHY INVESTIGATE THEORY?

For the purposes of this work we define ‘theory’ to mean *a broad class of concepts that aim to provide a structure for conceptual explanations or established practice*, and use such terms as ‘theories’, ‘models’, and ‘frameworks’ to describe particular manifestations of the general concept of theory. Examples encompassed by our definition include constructivism, cognitive load theory, Bloom’s taxonomy, grounded theories, phenomenographical outcome spaces, Simon’s classification system [25], and established pedagogical practices such as pair programming and contributing student pedagogies. A critical reader might think it impossible to draw a line between what should be included and what should not. Our basic goal, however, is to try to establish how the field builds on existing and emerging theoretical work. From this perspective, the inclusion or exclusion of an individual theory does not have a significant effect on the whole. On the other hand, applying very strict criteria for theory would not allow us to identify and list the many interesting and relevant connections between research works.

Examining the role of theory in educational design studies, diSessa and Cobb [5] write that “the importance of theory is completely uncontested” in many fields of science (p79), but is less widely accepted in educational research. They suggest that theory development is “critical for the long-term scientific health and practical power of design-related educational research” (p101).

We have not been able to find any analysis of how theories are being used in CER. However, such analysis has been carried in

mathematics education research, which has a much longer history than CER.

Niss [19] discusses the nature, origin, and foundation of theories in mathematics education, and identifies several types of theory: 1) overarching theories giving a general framework, such as social constructivism; 2) data-driven theories such as grounded theories; 3) theories that basically provide a terminology to use, such as process-object duality; and 4) theories offering a research method, such as phenomenography. It seems reasonable to suppose that these types of theory apply equally well to computing education research. In this paper, page limits constrain us to the first three types of theory; we hope that our analysis of research methods in CER will be published separately.

Niss identifies six different roles for theories. Theories can be used to *explain* observed phenomena or to predict an occurrence of a phenomenon. For example, cognitive load theory can use the concept of working memory explain the effects on students’ learning of task structure and presentation. The same theory can also provide *guidance for action or behavior* by reasoning about the principles by which learning and assessment tasks should be constructed.

Theories can provide a structured *set of lenses* to approach, observe, study or interpret the target of investigation. Good examples of this are phenomenography and variation theory, which allow us to investigate students’ experiences or understandings of a specific phenomenon and present them in a consistent way. Theories can also be used as a *safeguard against unscientific approaches*, by explicating underlying assumptions and choices, thus setting the framework for research. Finally, they can be used to *protect against attacks* from sceptical colleagues from other disciplines. By demonstrating that researchers share common theoretical understandings within the research community, rather than working with *ad hoc* concepts, frameworks and procedures, members of the community are better placed to argue about the quality of their work. It is thus clearly worthwhile to strive for a good understanding of the role of theories in CER.

Taking a different perspective, Pais et al. [20] discuss the difference between ‘How theories’, which focus on how to solve a practical problem, and ‘Why theories’, which try to explain what is happening behind the observed behavior – and which they emphasize should therefore play a greater role. In CER, Hundhausen et al. [11] analyzed a number of evaluation studies in algorithm visualization and concluded that constructivism could explain the observation that better learning results can be achieved by working actively with a visualization than by following it passively. Here constructivism is used as a Why-theory. Building on this meta-study, the engagement taxonomy [18] defines six different levels of engagement for student interaction and uses these levels to suggest how visualization tools should be designed. There is some empirical evidence to support the principles, but the taxonomy itself provides no clear explanation of why each level supports better learning than the levels below it. It would thus be a How theory.

There are more advantages to using a theory. Firstly, as Niss [19] has observed, they can provide a *terminology* to be used in discussion. This supports better communication of ideas and results between researchers, as they can use a common set of concepts and terms in preference to defining their own. This does

not rule out that the need to define new concepts and terms in an evolving field. Secondly, theories can suggest *hypotheses*, which can be tested empirically, such as the above-mentioned engagement taxonomy. Thirdly, they can provide better arguments for *interpreting empirical findings*, as in the case of the meta-study by Hundhausen et al [11]. Recent visualization research has therefore been investigating this complex interaction more closely, trying to either build evidence for the engagement taxonomy or refine it to better match new empirical findings.

Our current research is a first step to understanding the current role of theories in CER. In this paper, we focus on looking at how widely CER is building on existing theoretical work, where that work originates from, and to what extent CER is building on its own theoretical developments.

4. TMMCER CLASSIFICATION

In this work, we apply our TMMCER (Theories, Models and Methods in CER) classification system [16], which examines the theoretical/technical background and the research process evident in a paper. The system categorizes papers in seven dimensions: theoretical background, technical background, reference disciplines, research purpose, research framework, data source, and analysis method. In this paper we focus on just the two dimensions that are pertinent to the use of theories in CER.

Within the theoretical background dimension, **Theories, models, frameworks (TMF)** captures how the paper builds on previous theoretical research or established practice by *applying* or *extending* some TMF. We do not count or report methodological TMFs, such as phenomenography, as they are covered by other dimensions, and are outside the scope of this paper.

Reference discipline denotes the origin of a TMF by listing the field of study in which it has been developed: education, psychology, or engineering, for example. We do not list subfields, such as educational psychology. CER itself is listed if the TMF has its origin in that area. If several TMFs for a single paper build on the same discipline, we count it only once, as we are interested in finding out the share of CER papers that build on other disciplines.

5. RESEARCH METHOD

5.1 Data Pool

Our goal for this project is to build a broad view of the theoretical underpinnings of published computing education research. As the complete analysis of recent publications in all major publication venues was beyond our resources, we have had to make a pragmatic choice as to which venues to consider. We therefore chose to include Computer Science Education (CSEd) and ACM Transactions on Computing Education (TOCE), formerly published as Journal of Educational Resources in Computing (JERIC), as the most prominent journals that focus principally on computing education; and ICER, as a highly research-oriented conference [27] that accepted long papers (12 pages for most of the period we were studying, although this has now been reduced to eight pages). Long papers are important for our analysis, because they allow more discussion on theoretical issues and the research process.

There are many other journals and conferences that accept computing education research paper. But faced with a clear need to limit the scope of our research, we chose these three in the

belief that they offer the greatest concentration of computing education research, as opposed, for example, to computing education practice.

Having selected these three venues, we made three further decisions regarding the inclusion or exclusion of papers. We included all special issue papers from the journals, as special issues are a frequent and focal forum for the presentation of research. We excluded editorial papers and short summaries of other papers in the issue, as they generally do not present original research. And we excluded the four discussion papers from ICER 2011, as that year's conference clearly distinguished discussion papers from research papers and imposed a lower page limit on the former. All remaining papers could be categorized as papers presenting research. This gave a data pool of 308 papers from 2005-2011: 113 from CSEd, 98 from JERIC/TOCE, and 97 from ICER. We recognized that the transition from JERIC to TOCE in 2007-2008 might influence the results for this journal, and were interested in whether we would detect such an effect.

5.2 Research Process

The analysis concerning TMMCER categorization was carried out by ten researchers working in pairs. The data pool for 2005-2010 was divided evenly among the pairs, who classified the papers independently and then came to a consensus. The papers from 2011 were subsequently classified by the two first authors. While this paper reports only on TMFs and reference disciplines, we did classify the papers along all of the TMMCER dimensions.

In research of this nature, the inter-rater reliability of classifications results must always be established. We have addressed this matter in the following way. First, satisfactory inter-rater reliability between the same pairs of classifiers was established earlier in the project; full details of the reliability process and results are reported in an earlier paper [16]. Second, the one new project member, who had considerable experience in classifying papers using other schemes, was paired with one of the principal developers of our classification scheme to ensure optimal support for learning the new system. The leader in this pair took care that the interpretation of categories did not change from that used earlier in the project.

The process of identifying TMFs in the papers is not clear-cut, because we want to identify TMFs that are clearly used or extended in a paper, rather than just referred to as related work. For example, a paper might mention grounded theory in its literature review, but not apply grounded theory in the actual research. Such a paper cannot be said to be using grounded theory. We look for evidence that the theoretical constructs are used, for example, to guide research design, to formulate hypotheses, or to interpret previous or new results. Therefore we read the papers carefully to decide whether a TMF that was mentioned should be included or not. The main indicators for including a TMF were: 1) the paper had a (sub)section which presented its theoretical framework; 2) in the abstract, introduction or description of research design it was explained that the research was based on some specific theoretical framework; and 3) in the discussion section(s) it was explained that the results were interpreted using some theoretical framework. Investigation of the last two indicators included looking for citations and theory names in the text and whether the context of citations informed us that the work was using some theory. Citations that focused on building motivation for the

research or presenting parallel independent work were not counted. In many cases the analysis required negotiation between the classifiers to reach a consensus. Therefore the method of working in pairs turned out to be a clear advantage, despite the fact that the classification took a long time, several months, before all pairs had completed their efforts. There are definitely cases where the decision to include/exclude some TMF could easily have gone the other way. However, we are confident that the big picture is reasonably accurate, and that counts far more in this kind of research than whether there were 64, 65, or 66 papers in a particular category.

There were also occasional problems in determining reference disciplines. These were typically deduced from the forum in which the TMF was originally published, which in most cases provided a clear indication. However, we also had to make judgments between fields, as for reasons of clarity we listed only the principal fields. Thus, for example, TMFs in educational psychology may have been classified either in education or in psychology, as our expertise did not allow us to make a clear decision. Moreover, some TMFs might have clear origins that span multiple disciplines. Once again, however, the effect on the big picture is small.

6. RESULTS

6.1 Theories, Models and Frameworks

In 157 papers (51%) we identified at least one theory, model or framework (TMF) that had been used. We identified 314 instances of work based on TMFs, and 226 of these were *distinct* TMFs. The most common TMF was constructivism or some of its sub-theories (communal constructivism, constructionism, social constructivism, situated learning) (15 mentions), followed by some curricular framework (10 mentions, mostly different versions of ACM/IEEE Curriculum), the pair programming model (10), Bloom's taxonomy (7), and pedagogical patterns (6). Among other theories of learning and psychological theories, the most common were Bandura's self-efficacy theory (5), cognitive load theory (4), and schema theory (4). Among the 226 distinct TMFs, more than 150 could be identified with a designated name (such as Bloom's taxonomy or Kolb's experiential cycle), while the rest were generally references to individual papers. Almost 200 TMFs were identified only once in the whole data pool.

When considering the publication venues, a small majority of papers in both ICER (57%) and CSEd (57%) were based on some TMF, while the number for JERIC/TOCE was somewhat smaller (39%). This difference between the venues is significant (Pearson χ^2 , $p = 0.014$). We also looked at whether there were trends in the use of TMFs during the analysis period. Figure 1 shows the average share of papers that build on some TMF during the period. The total number of papers per year in all forums is fairly small (typically 20) and there is considerable yearly fluctuation in the results. We have tried to smooth this fluctuation by combining the results into three groups: the years 2005-2007 (135 papers), 2008-2009 (88 papers), and 2010-2011 (85 papers). The numbers of papers in these groups are far from equal, but they do serve to give a broad picture of possible trends. It seems that papers in JERIC/TOCE have increased their use of TMFs, suggesting that the transition from JERIC to TOCE had an effect on the characteristics of the papers, while there is no clear trend in the other venues.

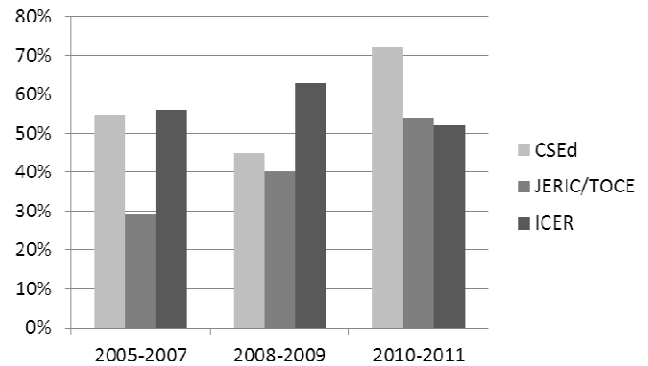


Figure 1: Proportions of papers at each venue that build on some TMF

Almost half of the papers do not appear to be based on any TMFs. This concurs with Wankat's findings from EER [31], although the numbers themselves cannot be directly compared, as Wankat's exact criteria for including TMFs were not given in his paper. Examining the 43% of the CSEd papers and the 43% of the ICER papers in which we had failed to find TMFs, we identified three types of paper. First, there were papers that formed new theories. These included various categorizations of data (e.g. phenomenographical outcome spaces), hypotheses that were formulated and possibly tested with the collected data, and new explanatory theories such as models or grounded theories. A second group of papers analyzed and discussed their data, perhaps presenting a new method of analysis, but did not present a finding with enough structure to be considered a theory in the sense being used in our analysis. Finally, there were papers including literature surveys, reports of technical contributions such as novel educational software or hardware, and reports of novel instructional methods. The first group of papers, those presenting new theories, formed a considerable share of the CSEd papers and a majority of the ICER papers, demonstrating that many new TMFs are being formulated within CER itself.

6.2 Reference Disciplines

For each paper with TMFs, we identified the reference disciplines from which those TMFs originated, counting only the *distinct* disciplines from each paper. This decision is based on the observation that only a very small share of the papers had more than one TMF from the same discipline, but these papers often had many of them. We decided that to reveal the general picture of how CER papers link to work in other disciplines, we should not give undue weight to such 'theory-rich' papers.

Although all of the analyzed papers are in computing education research, we listed CER as a reference discipline only where the identified TMF had been developed within CER; an example is contributing student pedagogy [9].

Narrowing the focus to papers that build on TMFs from other fields, we see that 20% (63) of these papers use some TMF from education, 13% (41) from psychology, and only 6% (20) from both. These are not large numbers. Almost half of the papers examined do not build on any theory, model, or framework, and 80% of the computing education papers in our data pool do not build on theoretical research from education. When interpreting these numbers it is worthwhile to recall that we deliberately adopted a loose definition of theory, and even then our findings

indicate low numbers of papers building on previous theoretical work.

Table 1 presents the percentages of reference disciplines for TMFs in each venue. The three disciplines of computing, education and psychology were referenced rather more than CER. The ‘others’ category includes medicine, philosophy, linguistics, management, and systems theory. Many papers had two or more reference disciplines, as they used TMFs from different areas. There are no significant differences between the venues (Pearson χ^2 , $p = 0.833$).

Table 1: Reference disciplines of TMFs as a percentage of the total reference disciplines from each venue

	CSEd	JERIC/TOCE	ICER	Total
Count	91	53	75	219
CER	15 %	17 %	17 %	16 %
Computing	27 %	25 %	20 %	24 %
Education	31 %	26 %	28 %	29 %
Psychology	18 %	15 %	23 %	19 %
Others	9 %	17 %	12 %	12 %

As we identified very few theories originating in CER (23 in all), we chose to look more closely at these. About half of them were categorization schemes or taxonomies, such as Simon’s system [25] or the engagement taxonomy for algorithm visualization [18]. The remainder included questionnaires, pedagogical practices such as contributing student pedagogy [9], and theoretical constructions such as Jadud’s EQ [12] and Hazzan’s ‘reducing abstraction’ framework [10]. While we have been using the word ‘theory’ more or less interchangeably with TMF, it is perhaps timely to recall that the full phrase is Theories, Models, and Frameworks; some of the TMFs mentioned above clearly fall into the second or third group rather than the first. The most common single TMF was the engagement taxonomy, used in three papers. As a whole, although the numbers are small, we conclude that there is some theoretical work in CER itself on which the field is building, and not just referencing as related work. As we noted in the previous section, there appears to be considerable new theoretical work in our data pool. This suggests that future research will have a wider pool of TMFs from CER on which to build. However, within our limited seven-year data pool, with such a small number of TMFs from CER, we are not yet able to identify such an increasing trend. We hope to see it in the future.

7. DISCUSSION

We have found clear evidence that the field of CER draws extensively on work from other disciplines. Over half of the TMFs originate in fields outside CER and computing, mostly in education and psychology. This is not a bad thing, but we do expect in the future to see more use of TMFs originating in CER.

We were surprised at the vast number of distinct TMFs that we identified, especially as we were counting only those that are really used or extended in the paper, not those that are just mentioned. However, it is in accord with Wankat’s findings in EER [31]. What might explain the large number? One obvious

factor is simply the richness of the CER field: people are investigating so many different things that many different TMFs need to be used. Moreover, different theories provide different points of view and can thus shed more light on the complex phenomena we are investigating. Another possible explanation could be pragmatic: because CER has created only few TMFs of its own, researchers must look elsewhere when seeking a solid foundation for their work. Most researchers in CER will be familiar with TMFs from the discipline of computing itself, but when those are not appropriate, they are left to study the broader literature on their own. The range of TMFs in the social sciences is large, so it is easy to imagine that researchers will adopt whatever TMFs they find that seem reasonably pertinent to their work. Moreover, many CER papers do not build clearly on any TMF. Thus it may be difficult to find work that is closely related to one’s own research topic and has a strong theoretical background, which would make it easier to use the same TMFs. Having read all of the papers in this data pool, we believe that if researchers were to explicitly list the theories and frameworks that they use, this would help other researchers to find work related to their own research, leading in turn to higher citation counts.

The richness brings advantages and disadvantages. It is a good sign that work from other research disciplines is widely used, and provides evidence that CER is truly cross-disciplinary and is actively adding to its theory base. Also the wealth of different theories provides more versatile views for the phenomena being investigated. On the other hand, this may also be a hindrance, as CER is building not on a small and stable theoretical base but on a large and dispersed one. There are mixed terminologies and ways of building arguments, as there will always be in cross-disciplinary work. Comparing one’s own results to those of others may not be easy when the results are interpreted according to different theoretical arguments. This may lead to a situation where knowledge in the field accumulates only in small isolated areas, which may be referenced as related work but which are not used as a foundation for new research.

This challenge has certainly been grasped as an opportunity in mathematics education research, as research combining different theoretical views may reveal novel understandings that are unlikely to emerge from single approaches [4]. Bikner-Ahsbabs and Prediger [3] discuss four different strategies for coping with the challenge. It is possible to *unify* theories that are developed locally to address similar phenomena or *integrate* theories by combining different approaches. Treating the same data set with different theories (*comparing and contrasting*) can reveal interesting differences and similarities in the theories while respecting the diversity. Finally, *networking* of theories is used when the previous strategies are systematically applied by first comparing and contrasting and then integrating new perspectives giving a progressively deeper understanding of the phenomenon. Research using any of these strategies is currently scarce in CER, but examples can be found [15, 28]. Of course this is not to diminish the value of papers that discuss the role of single theory in the CER context, such as those of Ben-Ari [1, 2]

A further finding of our analysis is that in nearly half of the papers we found no TMF. Our analysis cannot reveal why their authors have not based their work on any TMF, but as we have explained in the previous section, there can be some very good reasons. Therefore we are not concerned about each individual paper that

uses no TMF: our concern is more with the sheer proportion of such papers.

It must be noted that our analysis says nothing about how the TMFs are used in the papers we have examined, nor about the intrinsic worth of the TMFs we have found. These are highly important considerations, but are clearly well beyond the scope of this analysis. The merit of a simple count of TMFs lies not in what it says about any individual paper or the TMFs that it uses, but in the big picture that it presents of computing education research.

8. CONCLUSION AND IMPLICATIONS

We have carried out a comprehensive analysis of the papers published in two leading journals and the leading research-oriented conference. We have surveyed 308 publications from the seven years to 2011 to examine their theoretical underpinnings. The analysis has given us a deep look at CER, helping us to understand the many facets of the published research. Here we comment briefly on some of the main findings and present some recommendations for the whole field.

One of the goals of the work was to discover to what extent research in computing education is building on previous work, particularly theoretical work from other fields, because wide use of theoretical frameworks is one aspect of a maturing discipline. In mathematics education research this issue was widely discussed more than 20 years ago [29], and we hope that CER will consider it equally important.

Computing education research has been considered as a multi-disciplinary research field that combines computing with the human sciences, particularly education and psychology. We found clear support for this claim, as more than half of the papers in our data pool were based on one or more TMFs, more than half of those originating in disciplines outside CER and computing.

We also found TMFs from CER itself being used in other papers. These were mainly categorization schemes and taxonomies. Although they are not Why-theories [20] that can be used to explain observed phenomena from empirical work, they form a basis on which other research can better build. We hope to see more such work.

While much of the research in the field is somewhat insular, there is also a great and perhaps even excessive richness of the application of work from other fields. We suggest that the whole community of CER should emphasize building a better understanding of the role of TMFs in its research, not simply to learn more about specific theories but to discuss how the field can benefit from a broader use of theoretical work.

In order to address the challenge and opportunity of theory-richness within CER we propose the following.

- Some of the computing education conferences could host one- or two-day theoretical workshops in which a few carefully selected TMFs are discussed in some detail with invited experts in those theories, and a critical comparison is carried out of the literature and the findings from these approaches.
- Another way of organizing theoretical workshops could be that they solicit papers with a strong focus on applying different theories to bring about more visibility for the role of theory.

- We recommend the establishment of new research that purposefully combines different theoretical approaches. Journals in the field could solicit papers that compare, contrast or combine different theories in CER to be included in a special issue.
- When specifying review criteria in conferences and journals, the role of theory could be elaborated to clarify what expressions such as ‘solid theoretical framework’ actually imply. This includes giving some guidelines as to how the use of theoretical frameworks should be reported in the paper.
- As a community we should actively look at the achievements in our neighboring disciplines of mathematics, physics, science and engineering education research to learn about how theories are used in those contexts.

Through this work we have tried to capture the scope of theoretical underpinnings in work carried out in computing education research. This has increased our own understanding of the field as a whole, allowing us to make a number of observations and suggestions as to how the research tradition in the field could improve and mature. We hope that computing education research will continue to develop its own identifiable research tradition, shared methodologies, and some theories of its own, placing it on a more even footing with longer-established education research fields.

We have only skimmed the questions of the role and impact of theories in CER. Many questions remain to be addressed in our future work. What are the ‘homegrown’ TMFs in the field? How can they be characterized? Have they been validated in any way? How widely are they used in the CER literature outside our data pool? How and for what purposes are homegrown theories and borrowed theories from other disciplines used to support research (with reference to some of the theory-usage classifications from mathematics education research [3, 19, 20])? How are theories developed within CER? How is the acquired theoretical knowledge transferred into practical pedagogical content knowledge for computer science teachers? It is clear that much yet remains to be done.

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